MILITARY SPECIFICATION

CONTAINER, SHIPPING AND STORAGE, ALL UP ROUND

CNU-398/E

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and acceptance of the CNU-398/E all-up-round (AUR) container for the SIDEWINDER AIM-9 series missile, referred to herein as the container.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, New Jersey 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-101	Test Procedures for Packaging Materials

MILITARY

MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-648	Design Criteria for Specialized Shipping Containers

2.1.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

1297323	Sling
639AS2879	Dome Protector Assembly
1596AS100	Container, Shipping and Storage, CNU-398/E
1596AS104	Upper Shell Assembly
1596AS101	Cradle Assembly

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Container description. The container, which is of rectangular construction, includes an upper shell assembly and cradle assembly. The cradle assembly consists of a lower shell assembly and frame. A suspension system consisting of eight shock mounts is provided between the lower shell assembly and frame. Fasteners are positioned around the periphery to provide for rapid securing and opening. Wings and fins are carried in the lower shell assembly protected by dunnage. The container is airtight with a gasket providing proper closure seal. Missiles are held in place by a retainer assembly which can quickly be released for access to the missiles. A pressure relief valve is installed in the lower shell assembly. Skids, forklift pockets and lifting eyes are provided in the frame to facilitate handling.

3.1.1 Major components. The container shall consist of the following major components in accordance with Drawing 1596AS100:

a. Cradle assembly (Drawing 1596AS101).

b. Upper shell assembly (Drawing 1596AS104).

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.4.1 and 6.3).

3.3 General Characteristics. The container shall exhibit the following general characteristics:

a. The container shall be reusable and shall furnish transportation and storage protection for one to four SIDEWINDER all-up-round missiles (see 6.4.1) with storage included for one to four sets of wings and fins.

b. The container shall not be pressurized for shipment and storage.

3.4 Design and construction. Design and construction of the container shall be in accordance with Drawing 1596AS100 and as specified herein.

3.5 Performance requirements. The container shall meet all requirements specified herein when loaded with one to four AIM-9 series missiles, including wings and fins.

3.5.1 Leakage. When tested in accordance with FED-STD-101, method 5009, Pneumatic pressure technique, container leakage as indicated by a drop in pressure shall be not greater than 0.05 pound per square inch gage (psig) per hour (see 4.8.1).

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3.5.2 Shock. When the container is tested in accordance with the following specified methods of FED-STD-101, the container shall meet the acceptance criteria of MIL-STD-648, Mechanical shock.

a. Edgewise drop. The fully loaded container shall perform as specified herein when tested in accordance with FED-STD-101, method 5008 at -20 ±5 degrees Fahrenheit (°F) and 140 ±5°F (see 4.8.2).

b. Cornerwise drop. The fully loaded container shall perform as specified herein when tested in accordance with FED-STD-101, method 5005 at -20 ±5°F and 140 ±5°F (see 4.8.3).

c. Repetitive shock. The fully loaded container shall perform as specified herein when tested in accordance with MIL-STD-648, Repetitive shock test; and MIL-STD-648, Repetitive shock test, using a superimposed load equal to four fully loaded containers (see 4.8.4).

3.5.3 Vibration. When the fully loaded container is tested in accordance with FED-STD-101, method 5020, there shall be no evidence of physical or functional damage to container or contents, and there shall be no shifting of contents (see 4.8.5).

3.5.4. Impact. When the fully loaded container is tested in accordance with FED-STD-101, method 5012, or method 5023, at -20 ±5°F and 140 ±5°F; the container shall meet the acceptance criteria of MIL-STD-648, Mechanical shock (see 4.8.6).

3.5.5 Superimposed load. When the fully loaded container is tested in accordance with FED-STD-101, method 5016, there shall be no evidence of physical or functional damage to stacking features or other container features, or shifting of contents. Temporary or permanent deformation sufficient to result in any unsafe configuration or condition shall be cause for rejection of the container (see 4.8.7).

3.5.6. Mechanical handling. When the fully loaded container is tested in accordance with FED-STD-101, method 5011, Lifting and transporting by forklift truck, Hoisting with slings, Pushing, and Towing; there shall be no evidence of physical or functional damage to container handling features or other container features, or damage to or shifting of contents (see 4.8.8).

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3.5.7. Hoisting. When the container is suspended by its lifting eyes and tested in accordance with MIL-STD-648, Hoisting strength test, there shall be no evidence of structural or functional damage to the lifting eyes (see 4.8.9).

3.5.8 Stacked impact. When the fully loaded container, in unitized shipping configuration, is tested in accordance with MIL-STD-648, Impact test (stacked), there shall be no evidence of failure of the connecting structures which would permit any of the containers to become unattached (see 4.8.10).

3.5.9 Shipboard shock. When the fully loaded container is tested in accordance with MIL-STD-648, Shipboard shock test (eligibility), using a freefall drop height of 18.0 ±1.5 inches; there shall be no evidence of physical or functional damage to container or contents and there shall be no shifting of contents (see 4.8.11).

3.6. Reliability. Container shall be capable of 20 openings and closings with no evidence of physical or functional damage or evidence of wear that would require the replacement of any container component (see 4.11).

3.7. Configuration control. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor shall submit a supplemental sample to the procuring activity for appropriate inspections whenever a change is authorized by the procuring activity.

3.8 Government-loaned property. Unless otherwise specified in the contract or purchase order (see 6.2.1), the following property will be loaned to the contractor by the Government (see 6.5):

a. Four SIDEWINDER dummy rounds with wings and fins.

b. Four dome protectors (see Drawing 639AS2879).

c. Two slings (see Drawing 1297323).

3.9 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9, and the following. The container shall be constructed and finished in a manner to ensure compliance with all requirements specified herein. Particular attention shall be paid to neatness and thoroughness of assembly including construction of interface assemblies. The gasket cavity areas shall be free of excessive weld, weld splatter, voids, and excessive

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marks due to smoothing of welds. The coating and finish shall be even and neat, and the container shall be free of burrs and sharp edges. The standards of workmanship exhibited in any approved first article samples, subject to any qualifications stated in the Government notice of approval, shall establish the minimum workmanship standards for production items where such workmanship is not specifically covered by applicable standards (see 4.12).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Product verification inspection (see 4.6).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in 4.3.1.

4.3.1 Test conditions. Test conditions for inspections shall be the temperature, pressure and humidity conditions ambient to the testing facility.

4.4 First article inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the container shall be subjected to the first article inspection of Table I. The first article shall be inspected at a designated testing facility specified in the contract or purchase order

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TABLE I. First article, quality conformance, and product verification inspections.

Examination or test	Requirement paragraph	Method paragraph
a. First Article inspection		
Design and construction	3.4	4.9
Leakage	3.5.1	4.8.1
Edgewise drop	3.5.2a	4.8.2
Cornerwise drop	3.5.2b	4.8.3
Repetitive shock	3.5.2c	4.8.4
Vibration	3.5.3	4.8.5
Impact	3.5.4	4.8.6
Superimposed load	3.5.5	4.8.7
Mechanical handling	3.5.6	4.8.8
Hoisting	3.5.7	4.8.9
Stacked impact	3.5.8	4.8.10
Shipboard shock	3.5.9	4.8.11
Reliability	3.6	4.11
Workmanship	3.9	4.12
b. Quality conformance inspection		
Workmanship	3.9	4.12
Leakage	3.5.1	4.8.1
c. Product verification inspection		
Design and construction	3.4	4.9
Leakage	3.5.1	4.8.1
Edgewise drop	3.5.2a	4.8.2
Cornerwise drop	3.5.2b	4.8.3
Repetitive shock	3.5.2c	4.8.4
Vibration	3.5.3	4.8.5
Impact	3.5.4	4.8.6
Superimposed load	3.5.5	4.8.7
Mechanical handling	3.5.6	4.8.8
Hoisting	3.5.7	4.8.9
Stacked impact	3.5.8	4.8.10
Shipboard shock	3.5.9	4.8.11
Reliability	3.6	4.11
Workmanship	3.9	4.12

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(see 6.2.1). Failure of the first article sample to pass the first article inspection shall be cause for rejection of the first article. Any production of containers prior to first article approval shall be at the contractor's risk.

4.4.1 First article sample. The first article sample shall consist of two containers. The first article sample shall be representative of the methods, materials, processes and parts the contractor proposes to use in production of the container.

4.4.2 First article data. The contractor shall generate test reports documenting all tests of the first article inspection (see 6.2.2). Reports shall include complete details of the nature of the tests, test methods, test equipment, test results and result analyses.

4.5 Quality conformance inspection. The quality conformance inspection shall be as specified in Table I and shall be performed on 100 percent of the containers supplied under this specification. Failure of any container to pass the quality conformance inspection shall be cause for rejection of that container.

4.6. Product verification inspection. The product verification inspection shall be as specified in Table I.

4.6.1 Acceptance lot size. The lot size for each lot submitted for acceptance shall be as specified in the contract or purchase order (see 6.2.1), but shall be not greater than 100 containers.

4.6.1.1 Sample inspection lot size. The sample inspection lot size for each lot submitted for acceptance shall be one container which shall be selected at random.

4.6.2 Acceptance tests. Acceptance tests shall be performed by the contractor for each lot submitted for acceptance. The sample inspection lot shall be subjected to the product verification inspection specified in Table I. Failure of the sample of a sample inspection lot to pass any test of the product verification inspection shall be cause for rejection of that lot.

4.6.3 Acceptance data. The contractor shall generate test reports documenting all tests of the product verification inspection (see 6.2.2). Reports shall include complete details of the nature of the tests, test methods, test equipment, test results, and result analyses.

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4.7 Noncompliance. If a sample fails any first article or product verification inspection, the contractor shall notify the qualifying activity and the cognizant inspection activity of such failure. The contractor shall then generate a failure analysis to determine cause of the failure and take corrective action on the materials and processes, as warranted; and on all containers which can be corrected and which are considered subject to the same failure. After corrective action has been taken, inspection shall be repeated on one additional sample container (all tests and examinations, or the test which the original sample failed, at the option of the procuring activity (see 6.2.1)). Final acceptance and shipment shall be withheld until inspection has shown that the corrective action was successful.

4.8 Test methods.

4.8.1 Leakage. The container shall be tested in accordance with FED-STD-101, method 5009, Pneumatic pressure technique. Test pressure shall be 1.00 ±.05 psig and shall be applied for not less than two minutes. The container shall be pressurized by substituting a suitable fitting adapter in place of the container breather valve. This test shall be the first test performed and is required again after every test except the shipboard shock test. Upon completion of the test, the container shall meet the requirements of 3.5.1.

4.8.2 Edgewise drop. The container shall be fully loaded and inspected in accordance with method 5008 of FED-STD-101. Drop height shall be 17 ±1 inches. The test shall be performed using tri-axial accelerometers placed at the center of gravity of each missile and at a distance not greater than 12.0 ±.5 inches from the forward and aft end of each missile. The test shall be conducted at -20 ±5°F and again at 140 ±5°F. Upon completion of the test, the container shall meet the requirements of 3.5.2.

4.8.3 Cornerwise drop. The container shall be fully loaded and tested in accordance with method 5005 of FED-STD-101. Drop height shall be 17 ±1 inches. The test shall be performed using tri-axial accelerometers placed at the center of gravity of each missile and at a distance not greater than 12.0 ±.5 inches from the forward and aft end of each missile. The test shall be conducted at -20 ±5°F and again at 140 ±5°F. Upon completion of the test, the container shall meet the requirements of 3.5.2.

4.8.4 Repetitive shock. The container shall be fully loaded and tested in accordance with MIL-STD-648, Repetitive shock test; and MIL-STD-648, Repetitive shock test, using a superimposed load equal to four fully loaded containers. Upon completion of the test, the container shall meet the requirements of 3.5.2.

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4.8.5 Vibration. The container shall be fully loaded and tested in accordance with method 5020 of FED-STD-101. The test shall be conducted at ambient temperature utilizing tri-axial accelerometers placed at center of gravity of each missile and at a distance of not greater than 12.0 ±.5 inches from the forward and aft end of each missile. Upon completion of the test, the container shall meet the requirements of 3.5.3.

4.8.6 Impact. The fully loaded container shall be tested in accordance with method 5012 or method 5023 of FED-STD-101. The test shall be conducted at -20 ±5°F and 140 ±5°F and shall utilize tri-axial accelerometers placed at the center of gravity of each missile and at a distance of not greater than 12.0 ±.5 inches from the forward and aft end of each missile. Upon completion of the test, the container shall meet the requirements of 3.5.4.

4.8.7 Superimposed load. The fully loaded container shall be tested in accordance with method 5016 of FED-STD-101 with a superimposed load equal to four fully loaded containers. Upon completion of the test, the container shall meet the requirements of 3.5.5.

4.8.8 Mechanical handling. The fully loaded container shall be tested in accordance with FED-STD-101, method 5011, Lifting and transporting by forklift truck, Hoisting with slings, Pushing, and Towing. Upon completion of the test, the container shall meet the requirements of 3.5.6.

4.8.9 Hoisting. The container shall be tested in accordance with MIL-STD-648, Hoisting strength test. Upon completion of the test, the container shall meet the requirements of 3.5.7.

4.8.10 Stacked impact. The fully loaded container shall be tested in accordance with MIL-STD-648, Impact test (stacked). Upon completion of the test, the container shall meet the requirments of 3.5.8.

4.8.11 Shipboard shock. The fully loaded container shall be tested in accordance with MIL-STD-648, Shipboard shock test (eligibility), using a freefall drop height of 18.0 ±1.5 inches. Upon completion of the test, the container shall meet the requirements of 3.5.9.

4.9 Design and construction. To determine conformance to the drawing requirements of 3.4, the contractor shall perform the necessary visual and dimensional examinations to verify conformance to the dimensional and marking requirements of Drawing 1596AS100.

4.10 Materials certification. The contractor shall prepare certification that all materials are in accordance with the requirements of Drawing 1596AS100 and drawings referenced therein (see 6.2.2).

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4.11 Reliability. To determine conformance to 3.6, the container shall be opened and closed 20 times with no evidence of physical or functional damage or evidence of wear that would require the replacement of any container component.

4.12 Workmanship. To determine conformance to 3.9, the container shall be visually examined. Particular attention shall be paid to ensure that the gasket cavity is clean and free of excessive weld, weld spatter, voids and excessive marks due to smoothing of welds.

5. PACKAGING

5.1 Preservation. Dessicant in the amount of 1.2 units per cubic foot of volume shall be sealed in the container prior to shipment.

5.2 Packaging. Not applicable.

5.3 Packing. Unless otherwise specified in the contract or purchase order (see 6.2.1), containers shall be stacked for shipping in unitized loads of four each containers.

6. NOTES

6.1 Intended use. The container is intended for use in foreign and domestic shipment, transfer, and storage of up to four each SIDEWINDER AIM-9 series all-up-round missiles. Storage is provided in the container for corresponding wings and fins.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of specification.

b. Number of containers to be procured.

c. Submission of supplemental sample if different than specified (see 3.7).

d. Government-loaned property if different from 3.8.

e. First article inspection if different from 4.4.

f. Facility assigned to perform inspections (see 4.1 and 4.4).

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g. Facility assigned to perform first article inspection (see 4.4).

h. Size of acceptance lot, but not to exceed 100 containers (see 4.6.1).

i. Tests and examinations to be repeated in case of noncompliance (see 4.7).

j. Unitized shipping requirement if different from 5.3.

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specificaiton is cited in the following paragraphs.

Paragraph number	Data requirement title	Applicable DID number	Option
4.4.2	First Article Inspection Report	DI-T-4902	---
4.6.3	Production Inspection Reports	DI-T-4904	---
4.10	Report, Material Inspection System	UDI-T-23741A	---

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of two units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article sample.

6.4 Definitions.

6.4.1 All-up-round missile. Completely assembled missile less wings and fins.

6.5 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.8.

Preparing activity:

Navy - AS

(Project 8140-N647)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-85659(AS)	CONTAINER, SHIPPING AND STORAGE, ALL UP ROUND CNU-398/E

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording.

b. Recommended Wording.

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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